SCOPE-FD: A Client Selection Method in Federated Distillation for Massive-MIMO Systems

***Abstract*—Federated distillation lets edge devices train a shared model by exchanging only their predictions on a public dataset. That costs far less over a wireless link than sending weights. A base station can serve only a few of its devices per round, so it must choose which ones. Client selection is well studied for federated learning, where devices are ranked by a utility signal such as local loss. Federated distillation averages predictions rather than weights. That flattens the per-device signals, so the same rules do not carry over. Published systems for massive multiple-input multiple-output networks avoid the question by assuming that every device participates in every round. This paper proposes Server-aware Coverage with Over-round Participation Equalization, or SCOPE-FD, a deterministic selector for partial participation. It rotates devices by how far each has fallen behind a uniform participation target. Ties go to devices holding classes the server predicts poorly. The participation imbalance this produces is not merely bounded but exact. It follows from the pool size, the number served per round and the training horizon. An operator can therefore compute it before training starts. Composing that guarantee with the convergence bound of the underlying framework gives a rate in communication rounds. The rate holds for every device rather than on average. Accuracy is preserved rather than improved. Participation fairness is therefore constructible rather than merely achievable. It becomes something an operator can commit to in advance rather than verify afterwards.**

***Index Terms*—Federated distillation, client selection, massive MIMO, participation fairness, knowledge distillation, communication efficiency.**

***Impact Statement*—Wireless networks are starting to train shared artificial intelligence models across phones, wearables and vehicles. The data never leaves the device. Only a handful of devices can be served in each training round. Picking them at random means some devices contribute far more than others. Whose data shapes the model then becomes partly a matter of chance. This work replaces that lottery with a rotation. Its fairness follows from three quantities a network operator already knows. It can therefore be stated before a deployment is switched on rather than measured after it has run. Operators give up no accuracy in exchange. Artificial intelligence systems are increasingly asked to evidence fair treatment of the people they serve. Committing to a participation guarantee in advance, rather than reporting one in hindsight, changes what a deployment can promise its users and its regulators.**

I. Introduction

Federated Learning (FL) [1], [2] lets a population of devices train a shared model under the coordination of a central server without their data ever leaving them [3], [4]. In each round the selected clients train locally and upload the updated model parameters for aggregation. Uploading full model weights does not scale to modern architectures over a wireless edge, particularly when many devices share the same connectivity. Federated Distillation (FD) [5] avoids that cost by having each client upload only its output logits on a shared public dataset [6]. Such a logit-only exchange cuts the per-round communication overhead substantially and lets clients train heterogeneous local architectures [7]–[11].

The integration of FD with massive multiple-input multiple-output (mMIMO) wireless backbones is investigated in [12], where channel hardening and zero-forcing precoding handle the noisy uplink and downlink. The FedTSKD framework proposed there establishes a dynamic training schedule, an $\mathcal{O}(1/t)$ convergence bound, and a channel-aware aggregation rule for FD over mMIMO. It builds on the earlier wireless-FD line [13] and on wireless FL over fading channels [14], [15].

That work assumes full client participation, so all N clients transmit logits in every round, and the assumption is uniformly adopted across the FD literature [6]–[8], [11]. It breaks down whenever a deployment holds more devices than the scheduler can admit, runs under a per-round energy budget [16], or shares a contention-limited uplink. Each case forces the system to admit only $K \ll N$ clients per round. In FL, the question of which K out of N clients to admit per round has been studied from several directions, including resource-aware, importance- and channel-aware, bandit-driven, and submodular formulations [17]–[23], as reviewed in Section II-A. In FD the client selection problem has received scant attention, and the nearest prior work [24] selects public-dataset samples to distill on rather than clients to admit.

None of those FL selectors is designed for FD, for a structural reason. The FD aggregation rule is data-size-weighted logit averaging [6], [12], which flattens per-client quality differences once the selected set is fixed. The composition of that set, rather than any intra-client quality gradient, therefore drives the server model. FL behaves differently, since each local gradient enters the aggregate directly and ranking clients by training loss [21], [22] or gradient norm [25] produces a meaningful aggregate signal. The right question in FD client selection is therefore not how to score individual clients along a single utility axis. It is how to use every client's data uniformly across rounds, while nudging the within-round composition towards the regions of the label space where the server is currently uncertain.

A. Contributions

This paper proposes SCOPE-FD (Server-aware Coverage with Over-round Participation Equalization), a deterministic client selector for FD over mMIMO networks, built on the FedTSKD substrate of [12]. Existing FL selectors [17], [21]–[23] score clients by utility or by coverage but not both, while existing FD work [6], [12], [24] either assumes full participation or addresses it only along the data-sampling direction. The proposed score instead composes participation balance, server-side class under-prediction and per-round class coverage with non-negative coefficients held fixed across all configurations. To the best of the authors' knowledge it is the first deterministic fairness-guaranteed selector tailored to the data-size-weighted logit-averaging rule of FD under partial participation. The main contributions follow.

1) *Three-Term Deterministic Selector for FD.* We design a selector combining a participation-debt rotation term, a server-side under-prediction bonus and a class-coverage penalty, with the debt term primary and the other two breaking ties inside debt-equal cohorts. It carries zero trainable parameters, unlike the reinforcement-learning and neural-bandit FL selectors [21], [26] that maintain $\mathcal{O}(N \cdot K)$ trainable states, and fills K slots in $\mathcal{O}(KNC)$ time per round for C label classes.
2) *Participation-Fairness Guarantee.* We prove that the per-client cumulative participation counts differ by at most one at every round. When K divides N this sharpens to exact equality at the end of every $\lceil N/K \rceil$-round cycle. We further show that the cumulative coefficient after R rounds is not merely bounded but exactly $m(N - m)/(NKR)$, with $m = KR \bmod N$. This makes the $\mathcal{O}(1/R)$ bound tight and identifies $N \mid KR$ as the exact condition for it to vanish, so the fairness of a deployment can be computed in advance.
3) *Round-Indexed Convergence under Partial Participation.* The selector modifies only the composition of the per-round subset, leaving the local training, distillation and aggregation steps of FedTSKD [12] unchanged. Composing the per-client bound of [12] with the participation guarantee gives a rate of $\mathcal{O}(N/(KR))$ in communication rounds that holds for every client at every finite horizon rather than on average. It asks for nothing beyond the assumptions of [12], which we work through one at a time, and returns their $\mathcal{O}(1/R)$ result at full participation.
4) *Experimental Analysis.* Every quantity is a mean with a standard deviation over independent seeds. The datasets are Fashion-MNIST with non-independent and identically distributed (non-IID) Dirichlet partitions, MNIST and a spoken-digit corpus. The sweeps cover the cohort size K, the Dirichlet concentration α, the downlink signal-to-noise ratio (SNR), the pool size N and the two coefficients. The selector lowers the participation Gini coefficient by roughly an order of magnitude against uniform random in every configuration tested. On accuracy it ties at the headline setting and trails once the cohort reaches twenty clients, with the sparse-regime differences inside the seed variation. A four-way ablation separates each score term, and four published selectors ported into FD serve as comparisons.

Organization. The rest of this paper is organized as follows. Section II provides background on client selection in FL and FD. Section III details the system model and the FD aggregation rule under partial participation. The proposed SCOPE-FD algorithm is described in Section IV, followed by the theoretical analysis in Section V. Section VI presents the experimental results and Section VII concludes.

II. Background

A. Client Selection in Federated Learning

Existing FL client-selection work spans several directions. Resource-aware methods such as FedCS [17] admit the maximum number of clients that can train and upload within a per-round deadline. Importance- and channel-aware scheduling [18]–[20] jointly incorporates wireless link state and gradient magnitude into a per-client priority score. Bandit-driven online selectors [21], [22] treat per-client utility as the unknown reward of a multi-armed bandit. DivFL [23] constructs the per-round client subset by greedily maximizing a submodular function over client gradients. Complementary lines include adaptive resource-constrained selection [25] and the q-fair fairness objective [27]. Fairness has more recently been made the selection criterion itself rather than a constraint on it, as in the inclusive selection framework of [28], which scores clients on participation equity alongside utility. That line shares the objective of the present work, but it is formulated for FedAvg, where the selection signal enters the parameter average.

In all of the above the aggregation rule is FedAvg [1], where each selected update enters the aggregate directly, which is what makes a per-client quality score meaningful at the server. That property does not survive the FD aggregation rule.

Four published selectors serve as comparison points in Section VI, each ported into the FD pipeline. A client in FD uploads no gradient, so any selector whose native signal is a model update needs a substitute that FD produces. Two are available, namely the static label histogram and the per-round logit vector on the public set. DivFL [23] performs facility-location submodular maximization over client gradients and is ported by replacing the gradient with the public-set logit vector, which preserves the greedy procedure and changes only the representation. SubTrunc [29] performs truncated submodular maximization diversified by client loss and ports directly, since that loss is already computed every round. UnionFL [30] diversifies by historical participation, which maps onto the counts the selector already maintains. Oort [31] is a utility-based FL selector run without modification, which measures what happens when an FL rule moves to FD unchanged.

B. Federated Distillation Algorithms and Wireless Physical Layer Integration

The FD paradigm originates in [5], which proposed using knowledge distillation [32] to communicate model outputs rather than parameters. The semi-supervised variant DS-FL follows in [6], where a shared unlabeled public dataset carries the logit exchange and the server aggregates per-client logits by data-size-weighted averaging. Communication-efficient logit transmission has been further investigated in [7], where a soft-label quantization and delta-coding scheme is proposed, and in [8], where unlabeled auxiliary data is exploited through a certainty-weighted ensemble. The model-heterogeneous variant FedHe is introduced in [9], the clustered knowledge-transfer scheme of [10] extends FD to the personalization setting, and the multi-task formulation FedICT in [11] addresses multi-access edge computing scenarios.

Wireless-channel-aware FD is initiated in [13] for distributed edge learning over fading uplinks. The integration of FD with mMIMO backbones is then carried out in [12], where the FedTSKD framework is proposed together with its $\mathcal{O}(1/t)$ convergence guarantee. The FedTSKD work specializes the broader wireless FL over fading channels line of research [14], [15] to the FD paradigm. There the noise tolerance of the logit layer interacts favorably with mMIMO channel hardening, and zero-forcing precoding nulls inter-user interference.

All of the FD works above aggregate by data-size-weighted logit averaging, so once the per-round subset is fixed each client's contribution is governed only by its data-size weight and within-round scoring carries far less effect than in FL. None of them addresses how that subset should be constructed when only a strict subset can be admitted, since full participation is uniformly assumed.

C. Client Selection in Federated Distillation

The closest prior work is the active data sampling method of [24], which adaptively selects the public-dataset samples to distill on. That is orthogonal to client selection, since the rule operates on public samples rather than on the client population. Fairness-aware FL formulations such as the q-fair objective of [27] are a relevant baseline, but their constraint is defined on the FedAvg rule and does not transfer to logit averaging.

No existing work addresses three concerns together, namely partial participation in FD, the effect of data-size-weighted logit averaging on how the per-round subset shapes the server model, and the wireless-channel impairments of an mMIMO backhaul. The proposed SCOPE-FD algorithm is designed to address all three together, while preserving the convergence guarantee of FedTSKD [12].

Remark (Delta to [12]). The FedTSKD framework of [12], developed by the present authors' research group,[1] supplies the FD-over-mMIMO substrate. That substrate is the dynamic local-training schedule, the zero-forcing uplink and downlink processing, the data-size-weighted logit-averaging aggregation rule and the $\mathcal{O}(1/t)$ convergence analysis. SCOPE-FD inherits all of it without modification. What is new here is the partial-participation client-selection problem itself, which [12] does not address. The remaining contributions are the deterministic three-term score with non-negative coefficients held fixed across configurations, the participation Gini guarantee of Section V-A, and the round-indexed convergence corollaries of Section V-B.

III. System Model and Problem Formulation

A. Federated Distillation Setup

Consider an FD system where a base station (BS) equipped with N_{BS} antennas serves N single-antenna clients over an mMIMO wireless backbone. Client n holds a private local dataset $\mathcal{D}_n = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{|\mathcal{D}_n|}$, where $\boldsymbol{x}_i$ is the input sample and $y_i \in \{1, \ldots, C\}$ its class label, and no local dataset is shared with the BS at any stage. To capture the heterogeneity of edge deployments, the per-class proportions on each client are drawn from a Dirichlet distribution $\mathrm{Dir}(\alpha)$ over the label space [33], with smaller α giving more skewed partitions. A public unlabeled dataset $\mathcal{D}_p = \{\boldsymbol{x}_p\}_{p=1}^{|\mathcal{D}_p|}$ shared by the BS and all clients carries the per-client logit exchange [6]. Since only logits are transmitted, the clients are free to adopt heterogeneous local architectures [9].

The local loss at client n is given by

$$L_n(\boldsymbol{w}_n) = \frac{1}{|\mathcal{D}_n|} \sum_{(\boldsymbol{x}_i, y_i) \in \mathcal{D}_n} \ell(\boldsymbol{w}_n; \boldsymbol{x}_i, y_i), \tag{1}$$

where $\boldsymbol{w}_n$ denotes the local model parameters and $\ell(\cdot)$ is the per-sample cross-entropy. The server minimizes the data-size-weighted sum of local losses,

$$\min_{\{\boldsymbol{w}_n\}} F(\{\boldsymbol{w}_n\}) = \sum_{n=1}^{N} \frac{|\mathcal{D}_n|}{|\mathcal{D}|} L_n(\boldsymbol{w}_n), \tag{2}$$

where $|\mathcal{D}| = \sum_{n=1}^{N} |\mathcal{D}_n|$. Following [12], each communication round r runs three stages. These are local training, then uplink communication with server-side aggregation, then downlink broadcast with local distillation. The overall architecture, including the proposed selector at the cloud workstation, is summarized in Fig. 1.

In the local training stage, every client $n \in \mathcal{S}_r$ runs E_r steps of stochastic gradient descent (SGD) on its private dataset, where $\mathcal{S}_r$ denotes the set of clients selected in round r. The schedule E_r is set by the training configuration and never by the selector. It is held constant across rounds in every experiment reported here. The updated local parameters are then used to compute the per-sample logits $\boldsymbol{z}_{n,r}$ on the public dataset $\mathcal{D}_p$. After quantization and modulation, the logits are uploaded to the BS through the mMIMO uplink, and the received signal at the BS can be written as

$$\boldsymbol{Y}_r^{(\mathrm{UL})} = \sum_{n \in \mathcal{S}_r} \boldsymbol{H}_{n,r} \boldsymbol{Z}_{n,r}^T \sqrt{\frac{P_{\mathrm{UL}}}{N_D \sigma_z^2}} + \boldsymbol{\Omega}_r^{(\mathrm{UL})}, \tag{3}$$

where $\boldsymbol{Y}_r^{(\mathrm{UL})} \in \mathbb{C}^{N_{BS} \times N_D}$ stacks the N_D received symbol periods column-wise, $\boldsymbol{H}_{n,r} \in \mathbb{C}^{N_{BS} \times 1}$ is the uplink channel column vector between the single-antenna client n and the N_{BS}-antenna BS, $\boldsymbol{Z}_{n,r} \in \mathbb{C}^{N_D \times 1}$ is the complex-valued symbol representation of $\boldsymbol{z}_{n,r}$ obtained after quantization and modulation, so that $\boldsymbol{H}_{n,r}\boldsymbol{Z}_{n,r}^T$ is the rank-one $N_{BS} \times N_D$ matrix received from client n, P_{UL} is the uplink transmit power in linear scale, N_D is the number of complex data streams per logit vector, σ_z^2 is the per-symbol average power normalization in linear scale, and $\boldsymbol{\Omega}_r^{(\mathrm{UL})}$ is the receiver-side complex Gaussian noise. The scaling factor in (3) splits P_{UL} uniformly across the N_D streams and normalizes each entry of $\boldsymbol{Z}_{n,r}$ to unit average power. With zero-forcing (ZF) reception, the per-client logits at the BS can be recovered as

$$\hat{\boldsymbol{z}}_{n,r} = \boldsymbol{z}_{n,r} + \boldsymbol{\omega}_{n,r}^{(\mathrm{UL})}, \tag{4}$$

where the variance of the post-equalization noise $\boldsymbol{\omega}_{n,r}^{(\mathrm{UL})}$ is suppressed by a factor proportional to N_{BS}, owing to the channel hardening property of the mMIMO array [12], [15]. Recovering the K per-client streams by ZF requires the

[1]FedTSKD [12] is from the same research group as the present work and is treated here as an off-the-shelf substrate.

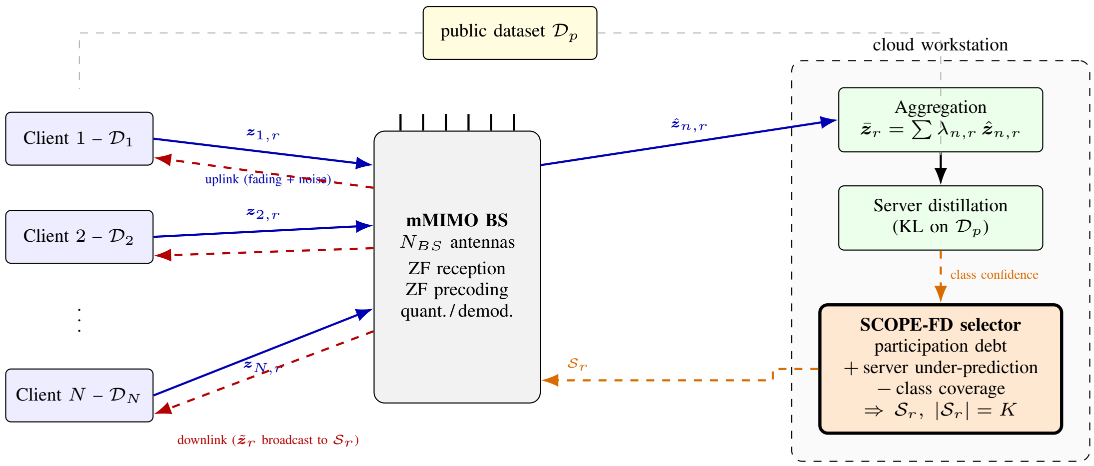

Fig. 1. Architecture of SCOPE-FD over an mMIMO backbone. Here $\boldsymbol{z}_{n,r}$, $\hat{\boldsymbol{z}}_{n,r}$ and $\tilde{\boldsymbol{z}}_r$ are the uplink logits, the ZF-equalized logits at the BS and the post-distillation logits broadcast downlink. The cloud workstation sits with the BS over a wired link and hosts the server model, the distillation step and the selector, so the only over-the-air segment is the BS-to-client interface. Solid blue arrows are uplink streams, dashed red the downlink broadcast and dashed orange the control path returning $\mathcal{S}_r$.

composite uplink channel formed by stacking $\{\boldsymbol{H}_{n,r}\}_{n\in\mathcal{S}_r}$ column-wise, an element of $\mathbb{C}^{N_{BS}\times K}$, to have full column rank, so the cohort size is bounded by the array, $K \le N_{BS}$. The per-round budget is therefore set by the spatial dimensions the BS can separate rather than by the size of the client pool.

B. FD Aggregation under Partial Participation

After dequantization and demodulation at the BS, the round-r aggregated logit is formed by data-size-weighted averaging over the selected clients, that is,

$$\bar{z}_r = \sum_{n\in\mathcal{S}_r} \lambda_{n,r}\,\hat{z}_{n,r}, \quad \lambda_{n,r} = \frac{|\mathcal{D}_n|}{\sum_{j\in\mathcal{S}_r}|\mathcal{D}_j|}. \tag{5}$$

The server-model parameters are then updated through a distillation step that minimizes the Kullback-Leibler divergence between the server-model output and $\bar{z}_r$ on the public dataset [32]. The post-distillation logits are subsequently quantized, modulated, precoded with a zero-forcing precoder, and broadcast to the selected clients over the mMIMO downlink. Each selected client then performs a local distillation step that updates its local model. As in the uplink, the downlink post-equalization noise variance is suppressed by the channel hardening of the mMIMO array [12].

Once $\mathcal{S}_r$ is fixed, $\bar{z}_r$ depends on the per-client logits only through the weights $\lambda_{n,r}$, so the over-round composition of $\mathcal{S}_r$ rather than any within-round utility drives the server model, which is what the proposed selector shapes.

C. Participation Fairness and Problem Statement

To quantify the per-client participation fairness, the Gini coefficient over the cumulative participation counts is adopted. After R rounds, let $n_i^{(R)}$ denote the total number of rounds in which client i has been selected. The participation Gini coefficient is given by

$$G^{(R)} = \frac{\sum_{i,j}\left|n_i^{(R)} - n_j^{(R)}\right|}{2N\sum_{i=1}^{N} n_i^{(R)}}. \tag{6}$$

A perfectly balanced selection policy yields $G^{(R)} = 0$, whereas uniform random selection yields $G^{(R)} > 0$, with the imbalance growing as the participation ratio K/N shrinks.

The goal of this work is to design a selection policy π that returns $\mathcal{S}_r \subset \{1,\dots,N\}$ with $|\mathcal{S}_r| = K$ in each round r, so as to minimize the FD convergence error while keeping the participation Gini coefficient small. Letting $\{\boldsymbol{w}_n^{(R)}\}$ denote the local model parameters after R rounds under policy π, the problem can be formulated as

$$\min_{\pi}\ \mathbb{E}\Big[F\Big(\{\boldsymbol{w}_n^{(R)}\}\Big)\Big] - F^*, \tag{7}$$

subject to three constraints. First, $|\mathcal{S}_r| = K$ for all r, which fixes the per-round communication budget. Second, the rule spends no uplink slots on selection signalling beyond the FedTSKD logit-exchange budget [12]. Third, the per-round computation at the BS is no greater than $\mathcal{O}(NKC)$ in the number of clients, slots and label classes, so the selector is light enough to run every round. Here, $F(\cdot)$ is the global objective defined in (2) and F^* is its optimal value.

IV. Proposed SCOPE-FD Algorithm

This section presents the proposed SCOPE-FD selector, which combines a participation-debt term, a server under-prediction bonus, and a class-coverage penalty into a single deterministic score, followed by the algorithm summary at the end of the section.

A. Three-Term Selection Score

For candidate client i at round r the SCOPE-FD score is

$$\mathrm{score}_i^{(r,k)} = \tilde{d}_i^{(r)} + \alpha_u\, b_i^{(r)} - \alpha_d\, p_i^{(r,k)}, \tag{8}$$

where $\tilde{d}_i^{(r)}$ is the participation-debt term, $b_i^{(r)}$ is the server under-prediction bonus, $p_i^{(r,k)}$ is the class-coverage penalty evaluated at the k-th greedy slot, and $\alpha_u, \alpha_d \ge 0$ are scalar coefficients that are kept fixed across all configurations. The selector fills the K slots of $\mathcal{S}_r$ greedily, in decreasing order of (8). After each pick the coverage vector that drives $p_i^{(r,k)}$ is updated, while the debt and bonus terms remain fixed within the round. The selector is a rotation, and deliberately so. Proposition 1 shows the debt ordering survives every coefficient pair with $\alpha_u + \alpha_d < 1$, so the complete score takes the K least-served clients and the information terms only break ties among them. The contribution is not that ordering, which round-robin also gives, but that this construction admits the closed-form guarantee of Section V-A, composes with the convergence bound of the substrate, and carries the information terms without disturbing either.

B. Participation-Debt Term

The participation-debt term is the primary signal and drives the over-round fairness of SCOPE-FD. Let $n_i^{(r-1)}$ be the cumulative participation count of client i over the first $r-1$ rounds. The uniform target after r rounds is $r \cdot K/N$, the count each client would hold had every previous selection been spread uniformly across the pool. The raw debt is the deviation of that target from the count through round $r-1$,

$$d_i^{(r)} = r \cdot \frac{K}{N} - n_i^{(r-1)}. \tag{9}$$

A positive $d_i^{(r)}$ means client i has been picked less often than the target and is a candidate for round r, while a negative value means over-picking. The raw debt is then mapped to the unit interval by min-max normalization across the pool,

$$\tilde{d}_i^{(r)} = \frac{d_i^{(r)} - \min_j d_j^{(r)}}{\max_j d_j^{(r)} - \min_j d_j^{(r)} + \varepsilon}, \tag{10}$$

where ε is a small positive constant added for numerical stability. The target in (9) is stated at r while the count runs to $r-1$. The offset is common to every client, so it cancels in (10) and neither the ordering nor Proposition 1 depends on it. Under (10) the most under-picked client of the round receives $\tilde{d}_i^{(r)} = 1/(1+\varepsilon)$ and the most over-picked receives 0, so the term spans the full $[0,1]$ range whatever the magnitude of the raw debt. When every count is equal the numerator vanishes for all clients and the term is uniformly zero, in which case the two information terms alone order the round. The history in (9) is already kept at the BS as a by-product of scheduling and needs no client-side signalling.

C. Server Under-Prediction Bonus

The server under-prediction bonus is a secondary signal that tilts the selection towards clients whose local data covers classes the server model currently under-predicts on the public dataset. Let $\boldsymbol{u}^{(r)} \in \mathbb{R}^C$ denote the per-class under-prediction vector at round r, whose c-th entry is one minus the average softmax mass the server places on class c across $\mathcal{D}_p$, namely

$$u_c^{(r)} = 1 - \frac{1}{|\mathcal{D}_p|} \sum_{p=1}^{|\mathcal{D}_p|} \mathrm{softmax}_c(g(\boldsymbol{x}_p; \boldsymbol{\theta}_r)), \tag{11}$$

where $\boldsymbol{\theta}_r$ denotes the server-model parameters at the start of round r (i.e., after the round-$(r-1)$ distillation step in [12, Algorithm 1]) and $g(\boldsymbol{x}_p; \boldsymbol{\theta}_r) \in \mathbb{R}^C$ is the corresponding server-model logit on the public sample $\boldsymbol{x}_p$. A high $u_c^{(r)}$ means class c is rarely predicted by the server on $\mathcal{D}_p$, whether from genuine uncertainty, class bias or a skewed public-data prior, and the bonus does not distinguish these causes. Letting $\boldsymbol{h}_i \in \mathbb{R}^C$ denote the L1-normalized label histogram of client i, the bonus is given by

$$b_i^{(r)} = \langle \boldsymbol{h}_i, \boldsymbol{u}^{(r)} \rangle = \sum_{c=1}^{C} h_{i,c}\, u_c^{(r)}. \tag{12}$$

The score is therefore biased towards clients whose histograms overlap the under-predicted classes, targeting label-space regions the server has not yet learnt to recover. The vector $\boldsymbol{u}^{(r)}$ is computed at the cloud workstation from softmax outputs the distillation step already produces, so no extra uplink is incurred. Each $\boldsymbol{h}_i$ is uploaded once at session start, and its $\mathcal{O}(C)$ overhead is dominated by one round of logit exchange.

D. Class-Coverage Penalty

The class-coverage penalty discourages picking a client whose label distribution overlaps heavily with the classes already covered by the partial selection $\mathcal{S}_r^{(<k)}$ at the k-th greedy slot. Let $\boldsymbol{c}^{(r,k)} \in [0,1]^C$ denote the per-class coverage vector at the start of slot k, defined as

$$c_j^{(r,k)} = \min\left(1, \sum_{n \in \mathcal{S}_r^{(<k)}} h_{n,j}\right), \tag{13}$$

which saturates at unity once class j has been fully covered by the partial selection. The saturation keeps each entry of $\boldsymbol{c}^{(r,k)}$ in $[0,1]$ and prevents over-counting classes already represented. The coverage penalty for candidate i at slot k is the inner product

$$p_i^{(r,k)} = \langle \boldsymbol{h}_i, \boldsymbol{c}^{(r,k)} \rangle. \tag{14}$$

At the first slot of each round, $\boldsymbol{c}^{(r,1)} = \boldsymbol{0}$ and the penalty is uniformly zero. At subsequent slots, the penalty grows for clients whose histograms overlap with the already-selected classes, so the within-round selection naturally diversifies across the label space.

Privacy consideration. Both (12) and (14) read the per-client label histogram $\boldsymbol{h}_i$, which discloses the per-class distribution of the local dataset to the server. Two remedies are compatible with the proposed selector and both are evaluated in Section VI-H. The first perturbs the histogram before upload. Since $\boldsymbol{h}_i$ is L1-normalized, replacing one client's dataset moves it by at most 2 in L1, so Laplace noise of scale $2/\varepsilon_{\mathrm{dp}}$ per entry followed by renormalization gives that client $(\varepsilon_{\mathrm{dp}}, 0)$-differential privacy. The second drops the upload, replacing $\boldsymbol{h}_i$ by a server-side surrogate formed from the softmax-averaged public-set logits of (5). It is defined only after a client has been selected once, so the pool average stands in until then, which leaves the debt ordering and Proposition 1 untouched. Being built from submitted logits, it narrows the disclosure rather than ending it.

Choice of the coefficients. The coefficients are held fixed at $(\alpha_u, \alpha_d) = (0.3, 0.1)$ throughout Section VI and are not tuned

Algorithm 1 SCOPE-FD Selector at Round r

Require: Round index r, slot count K, label histograms $\{\boldsymbol{h}_i\}$, participation counts $\{n_i^{(r-1)}\}$;
Require: Under-prediction vector $\boldsymbol{u}^{(r)}$, coefficients α_u, α_d
Ensure: Selected subset $\mathcal{S}_r$ with $|\mathcal{S}_r| = K$
Compute $\tilde{d}_i^{(r)}$ for all i via (9)–(10)
Compute $b_i^{(r)}$ for all i via (12)
Initialize $\mathcal{S}_r \leftarrow \emptyset$ and $\boldsymbol{c}^{(r,1)} \leftarrow \boldsymbol{0}$
for $k = 1$ to K **do**
  Compute $p_i^{(r,k)}$ for all $i \notin \mathcal{S}_r$ via (14)
  $i^* \leftarrow \arg\max_{i \notin \mathcal{S}_r} \tilde{d}_i^{(r)} + \alpha_u b_i^{(r)} - \alpha_d p_i^{(r,k)}$ (ties broken by smallest client index)
  $\mathcal{S}_r \leftarrow \mathcal{S}_r \cup \{i^*\}$
  Update $\boldsymbol{c}^{(r,k+1)}$ via (13)
end for
return $\mathcal{S}_r$

for accuracy. They are chosen so the debt term stays dominant, which is the condition Proposition 1 requires. The normalized debt in (10) changes by a full unit between a client one round behind and a client that is level. The bonus in (12) and the penalty in (14) are both bounded in $[0, 1]$, so their combined influence is at most $\alpha_u + \alpha_d$. Any setting with $\alpha_u + \alpha_d < 1$ therefore leaves the debt ordering intact while letting the two information terms break ties inside a debt-equal cohort.

This reasoning is verified empirically in Section VI-C, where a grid over α_u and α_d shows that the participation Gini coefficient is unchanged across every admissible setting and that the final accuracy varies by little more than one percentage point over the whole grid.

E. Algorithm Summary and Complexity

Algorithm 1 summarizes the proposed SCOPE-FD selector at the BS. In the first round, all participation counts are zero and the debt is uniform, so the selection reduces to picking K clients by the bonus minus the penalty alone. From the second round onward, the debt term dominates the score and drives the over-round rotation, with the bonus and penalty acting as tie-breakers within debt-equal cohorts.

The per-round computational cost at the BS is $\mathcal{O}(NKC)$, since the debt and bonus terms cost $\mathcal{O}(NC)$, and the greedy outer loop runs K iterations with $\mathcal{O}(NC)$ per iteration for the coverage penalty. The memory footprint is $\mathcal{O}(NC)$, dominated by the per-client label histograms. Both complexities are linear in N for fixed K and C, satisfying the third constraint in Section III-C.

V. Theoretical Analysis

The participation-balance guarantee is established first. It is then composed with the per-client bound of FedTSKD [12] to give a rate in rounds under partial participation.

A. Participation-Balance Guarantee

Proposition 1 (Bounded Participation Spread). Under the SCOPE-FD selector with non-negative coefficients α_u, α_d such that $\alpha_u + \alpha_d < 1$, which is satisfied by the calibration $(0.3, 0.1)$ used in this paper, the per-client cumulative participation counts after any R rounds satisfy

$$\left| n_i^{(R)} - n_j^{(R)} \right| \leq 1, \qquad \forall\, i, j \in \{1, \ldots, N\}, \tag{15}$$

that is, no two clients differ by more than one in cumulative participation. The Gini coefficient is then bounded by

$$G^{(R)} \leq \frac{N}{4KR}, \tag{16}$$

which decays as $\mathcal{O}(1/R)$ for fixed N and K.

Proof sketch. The argument proceeds by induction on the round index r. The base case $r = 1$ is trivial since all participation counts are zero. For the inductive step, assume $|n_i^{(r-1)} - n_j^{(r-1)}| \leq 1$ at the start of round r and let $n_{\min} = \min_i n_i^{(r-1)}$, so the counts lie in $\{n_{\min}, n_{\min}+1\}$. Call the two cohorts *under-picked* and *over-picked*, respectively. If all counts are equal the step is trivial, since (10) is then flat and any K picks leave the counts in $\{n_{\min}, n_{\min}+1\}$. Assume therefore that both cohorts are non-empty. The raw debts (9) take exactly two values that differ by one. The min-max normalization (10) therefore sends the under-picked cohort to $\tilde{d}_i^{(r)} = 1/(1+\varepsilon)$ and the over-picked cohort to $\tilde{d}_i^{(r)} = 0$. Here ε is small enough that $1/(1+\varepsilon) > \alpha_u + \alpha_d$, which the calibration $(0.3, 0.1)$ satisfies by orders of magnitude for any standard numerical floor. Since $b_i^{(r)}, p_i^{(r,k)} \in [0, 1]$, the score gap between any under-picked and any over-picked client is at least $1/(1+\varepsilon) - (\alpha_u + \alpha_d) > 0$. Because $\tilde{d}_i^{(r)}$ is held fixed within the round and the bound on $p_i^{(r,k)}$ is slot-independent, the gap holds uniformly across all K greedy slots, so the greedy filling picks all under-picked clients before any over-picked one. If the under-picked cohort has size $\geq K$, all K picks come from it and the round-end counts again lie in $\{n_{\min}, n_{\min}+1\}$. Otherwise every under-picked client is taken together with a subset of the over-picked cohort, leaving the former at $n_{\min}+1$, the picked over-picked clients at $n_{\min}+2$ and the unpicked ones at $n_{\min}+1$, so the counts lie in $\{n_{\min}+1, n_{\min}+2\}$. Either way the spread is one and (15) holds, completing the induction. For the Gini bound, if a clients end at $n_{\min}$ and $N - a$ at $n_{\min}+1$, the numerator of (6) equals $2a(N-a) \leq N^2/2$ and the denominator equals $2NKR$, which yields (16). ■

When N is an integer multiple of K, the cycle length $\lceil N/K \rceil = N/K$ corresponds to exactly N selection slots, and (15) sharpens to equality of all participation counts at the end of each cycle. The participation Gini coefficient is then identically zero over every N/K-round window, irrespective of the channel state and the Dirichlet heterogeneity level α, starting from the standard zero-initialized participation counts.

The proof above discards information that is worth keeping. The step that produces (16) replaces the exact numerator $2a(N-a)$ by its maximum $N^2/2$. Retaining it gives the cumulative Gini coefficient in closed form. After R rounds the selector has filled KR slots, so the counts take $\lfloor KR/N \rfloor + 1$ for exactly $m = KR \bmod N$ clients and $\lfloor KR/N \rfloor$ for the other $N - m$. Substituting $a = N - m$ gives

$$G^{(R)} = \frac{m\,(N-m)}{NKR}, \qquad m = KR \bmod N. \tag{17}$$

Three consequences follow, each checked against measurement in Section VI-M. First, the bound (16) is tight rather than merely sufficient, since $m(N-m)$ is maximized at $m = N/2$, where it recovers (16) with equality for even N. Second, the coefficient vanishes exactly when N divides KR, a condition on the horizon as well as on the pool and the cohort, so the divisibility of N by K is neither necessary nor sufficient. At the headline setting $N = 30$, $K = 5$, $R = 100$ the cohort divides the pool, yet KR leaves a remainder of 20 and the coefficient settles at 1.33%. At $N = 50$ with $K = 3$ the cohort does not divide the pool, yet $KR = 300$ is an exact multiple of 50 and the coefficient is zero. Third, the decay in R is not smooth. Its envelope falls as $\mathcal{O}(1/R)$ while the value oscillates with m, touching zero whenever the horizon completes a whole number of rotations.

The divisibility condition $K \mid N$ is needed only for the per-window statement, where each cycle of $\lceil N/K \rceil$ rounds serves every client once. The bound (15) is proved without it and holds at every round, so neither $K \nmid N$ nor an offset window breaks it. Both cases are examined in Section VI-K.

Proposition 1 governs participation balance only. The bonus and penalty terms enter the score strictly within the debt-equal cohort, so they do not affect the spread bound (15). They refine the composition of $\mathcal{S}_r$ within a round, not its over-round counts.

B. *Convergence under Deterministic Partial Participation*

The selector changes which clients enter $\mathcal{S}_r$ and nothing else, so the local training, the server-side aggregation and the distillation steps stay those of FedTSKD [12]. Three properties of that work fix what follows. Its Algorithm 1 admits all N clients in every round and its Section VI adopts full participation by design, so it carries no client-selection mechanism. Its Assumptions 1–4 are smoothness and strong convexity of the local training and distillation losses with bounded variance and gradient norm, and none refers to $\mathcal{S}_r$. The clause *uniformly and independently* in Assumption 3 describes the minibatch $\mathcal{D}_{n,t} \subset \mathcal{D}_n$ drawn inside client n rather than how clients are drawn. Its proof of Theorem 1 tracks one client through its own weight sequence and no step reads another client's state, which follows from the first property, since an algorithm with no selection mechanism leaves its proof nothing to condition on.

Two consequences follow and they carry different weight. The per-client bound of [12, Theorem 1],

$$\mathbb{E}[L_n(\boldsymbol{w}_{n,t+1})] - L_n^* \le \frac{\psi}{2} \cdot \frac{\nu_n}{t+1+\beta_1}, \tag{18}$$

carries over to SCOPE-FD unaltered, with ψ the smoothness constant of the global objective, ν_n the per-client local-gradient variance budget and β_1 the regularization offset of that theorem. That much needs no argument, because (18) was never conditioned on how clients are chosen. The substantive point is the index t, which counts the steps client n has actually taken. Under full participation it moves with the round counter. Under partial participation it advances only in the rounds where that client is selected, so (18) alone bounds nothing in terms of the communication budget R. Proposition 1 closes that gap for every client at once.

Corollary 1 (Per-Client Rate in Rounds). Let E_d be the number of distillation steps per round, let $E_{\min} = \min_{r \le R} E_r > 0$, which equals the constant E_r of every experiment reported here, and write $\Gamma_R = (E_{\min} + E_d)(KR/N - 1) + 1 + \beta_1$. Under Assumptions 1–4 of [12] and $\alpha_u + \alpha_d < 1$,

$$\mathbb{E}\big[L_n(\boldsymbol{w}_n^{(R)})\big] - L_n^* \le \frac{\psi\, \nu_n}{2\,\Gamma_R} \tag{19}$$

for every R large enough to make Γ_R positive, so the per-client error falls as $\mathcal{O}(N/(KR))$.

Proof. Only the clients in $\mathcal{S}_r$ update in round r, so the clock of client n advances by $E_r + E_d \ge E_{\min} + E_d$ steps on each round where it is selected and by nothing on the others. After R rounds it stands at $E_{\min} + E_d$ times that client's participation count or above, and the counting argument behind (17) puts every count at $\lfloor KR/N \rfloor$ or above, which is at least $KR/N - 1$. Substituting both facts into (18) gives the claim. ■

The word *every* is what Proposition 1 contributes. A policy that balances participation only in expectation cannot make the same statement at a finite horizon, since one realization may leave a client far below KR/N however well the average behaves. Uniform random selection is such a policy.

Corollary 2 (Global Objective in Rounds). Under the same conditions, and with $\bar{\nu} = \sum_n \lambda_n \nu_n$ for $\lambda_n = |\mathcal{D}_n|/|\mathcal{D}|$,

$$\mathbb{E}\big[F(\{\boldsymbol{w}_n^{(R)}\})\big] - F^* \le \frac{\psi\, \bar{\nu}}{2\,\Gamma_R}, \tag{20}$$

which is again $\mathcal{O}(N/(KR))$.

Proof. Only ν_n depends on n in (19), since Γ_R does not, so weighting by λ_n and summing gives (20). The weights here are the global λ_n of (2) rather than the within-round $\lambda_{n,r}$ of (5), which is what the objective is defined against. The objective (2) is the λ_n-weighted sum of the local losses and [12] poses N separate per-client minimizations, so $F^* = \sum_n \lambda_n L_n^*$. ■

Setting $K = N$ recovers $\mathcal{O}(1/R)$, the rate reported in [12] under full participation, which is a consistency check rather than a new result. Away from that point the cost of partial participation appears as the factor N/K and nowhere else. The sweep over K in Section VI-E varies exactly that ratio.

Which assumptions are used. Assumptions 1 and 2 of [12] are properties of the loss functions and SCOPE-FD changes no loss function. Assumptions 3 and 4 add bounded variance and gradient norm under within-client minibatch sampling, which the selector never touches, and the aggregation reads $\mathcal{S}_r$ only through (5), used unaltered. There is no client-sampling assumption in [12] for a deterministic rotation to satisfy or to violate, so Corollary 2 is a consequence of Theorem 1 of that work composed with Proposition 1 rather than a parallel result resting on added conditions. What the corollaries do not give is a rate that beats uniform random selection in order, since both are $\mathcal{O}(N/(KR))$. The gain of the rotation is that the bound holds for each client at every finite R instead of only on average, which is the distinction Proposition 1 already draws for the counts themselves. Two limits come with the substrate. Since (18) measures each local loss against its own optimum and (2) is a separable sum of those losses, the corollaries do not by themselves separate federated from isolated local training, and the server-model accuracy of Section VI is not a quantity they bound. They give the round-indexed rate of the substrate under a deterministic rotation and nothing past it.

VI. Experimental Results

The study below evaluates SCOPE-FD as a fairness-first selector, giving a Pareto improvement on the accuracy and participation-Gini pair rather than a uniform accuracy gain.

A. Experimental Setup

The proposed SCOPE-FD selector is integrated into the FedTSKD framework [12] and benchmarked against uniform random selection and four ported selectors on Fashion-MNIST (FMNIST). The common simulation parameters are listed in Table I and mirror those of [12] where applicable, with the partial-participation K and the selection-side coefficients α_u, α_d introduced by the present work.

Reporting protocol. Every reported quantity is a mean with a standard deviation over independent seeds. The headline configuration and the selector comparison use five seeds, the parameter sweeps three. The canonical value there is 71.21 ± 0.99 over the five headline seeds. Three-seed families report 71.75 ± 0.81, which is what the five-seed runs return on that seed subset, so the difference is the seed set and not the configuration, and the 71.99 ± 0.63 of the privacy study sits 0.24 points from it as run-to-run variation. Comparisons are made within a family on matched seeds. Accuracy comparisons use a paired Wilcoxon signed-rank test over matched seeds, so the partition and the channel realization are fixed within each pair. With five seeds the smallest attainable two-sided p is 0.0625, so the test supports statements of equivalence rather than significance. Convergence speed is the number of rounds to reach a fixed absolute accuracy threshold, since a threshold relative to each method's own ceiling is not comparable across methods. Each client holds a private partition with Dirichlet concentration $\alpha = 0.5$, and clients adopt heterogeneous local convolutional neural network (CNN) architectures from the FD-CNN1/2/3 pool [9]. The public set used for logit exchange [6] is 2000 unlabeled samples drawn once from the held-out split of the private dataset under a fixed sampling seed. Only the inputs are used, since the aggregation rule (5) consumes public-set logits and never reads a public-set label. They come from the same held-out split on which accuracy is measured and are not removed from it, so the public inputs are a subset of the evaluation inputs. No label is exposed and every selector sees the identical set, so the comparison is unaffected, and Section VI-G reports a disjoint-corpus control drawn from MNIST and treats the public set as a variable. The mMIMO BS is equipped with $N_{BS} = 64$ antennas. The channel model is enabled only in Sections VI-F and VI-J, with the uplink SNR at -8 dB and the downlink SNR swept over $\{-10, -15, -20, -25, -30\}$ dB. The logit exchange is error-free elsewhere, so accuracies are not comparable across that boundary. The distillation temperature is fixed at $T = 1.0$ to match the underlying FedTSKD framework [12]. FMNIST is the primary benchmark because it leaves enough headroom for the selection rule to be observable across both sweeps. MNIST and the FSDD test whether the conclusions carry beyond it, in Section VI-L. SCOPE-FD is paired against uniform random on the same partition, channel realization and seed, so any difference is attributable to the selection rule alone.

TABLE I
Common simulation setup, mirroring [12].

Parameter	Value
BS antennas N_{BS}	64
Channel model	disabled by default, enabled in Sections VI-F and VI-J
Uplink SNR	-8 dB when the channel model is enabled
Downlink SNR	swept over $\{-10, -15, -20, -25, -30\}$ dB when enabled
Communication rounds R	100
Private datasets	FMNIST (primary), MNIST, Free Spoken Digit Dataset (FSDD)
Public dataset	unlabeled held-out split of the private set, replaced by MNIST in the sensitivity study
Public-set size	2000 samples, swept over $\{100, 500, 2000\}$
Client pool N	30 default, swept over $\{30, 47, 50, 53, 100, 200\}$
Cohort size K	5 default, swept over $\{1, 3, 5, 6, 7, 10, 20, 40\}$
Dirichlet α	0.5 default, swept over $\{0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 5.0\}$ plus IID
Client architectures	FD-CNN1 / FD-CNN2 / FD-CNN3
Optimizer	SGD (local), Adam (server distillation)
Learning rate	10^{-2} (local), 10^{-3} (distillation)
Distillation temperature	1.0
Logit quantization	8-bit
SCOPE-FD coefficients	$\alpha_u = 0.3,\ \alpha_d = 0.1$
Seeds	5 seeds headline, 3 seeds sweeps

B. Score Terms and Comparison with Ported Selectors

Two questions are settled at the headline configuration $N = 30$, $K = 5$, $\alpha = 0.5$ over five seeds. The first is how much each score term contributes, answered by the four variants in the upper block of Table II. The second is how the selector compares against the published methods ported into FD in Section II-A, which is the lower block. The complete score returns 71.21 ± 0.99 in both, so the blocks share an anchor.

The participation-debt term supplies the entire fairness result. All four variants reach a Gini coefficient of 1.33%, the value (17) specifies here, and removing either information term leaves it unchanged, as Proposition 1 predicts since those terms act only inside a debt-equal cohort. The four are also indistinguishable in final accuracy, with standard deviations overlapping in every pair. The information terms show up only in convergence speed, where the complete score reaches 60% absolute accuracy in 13.0 rounds against 14.0 for the debt term alone. The separation against uniform random is larger, at 13.0 rounds against 19.0. A stricter 70% threshold is not reached by every seed within the 100-round budget, so it is read as an attainment count. Four of five seeds reach it for the proposed selector and for uniform random, one

TABLE II

SCORE-TERM ABLATION (UPPER) AND SELECTOR COMPARISON (LOWER). FMNIST, $N = 30$, $K = 5$, $\alpha = 0.5$, FIVE SEEDS, MEAN $\pm$ STANDARD DEVIATION. EVERY SEED EXCEPT OORT'S REACHES THE 60% THRESHOLD, SO THAT COLUMN AVERAGES THE FULL SET. ACCURACY IS EVALUATED EVERY FIVE ROUNDS, SO THAT COLUMN TAKES VALUES IN $\{11, 16, 21\}$ AND ITS DISPERSION IS QUANTIZED.

Selector	Accuracy (%)	Gini (%)	Rounds to 60%
Debt only	71.26 ± 1.14	1.33	14.0 ± 2.7
Debt + under-pred.	70.55 ± 1.19	1.33	14.0 ± 2.7
Debt + coverage	71.56 ± 0.98	1.33	13.0 ± 2.7
Complete SCOPE-FD	71.21 ± 0.99	1.33	13.0 ± 2.7
UnionFL [30]	71.18 ± 1.40	1.33	13.0 ± 2.7
Uniform random	70.99 ± 1.38	12.49	19.0 ± 2.7
DivFL [23]	69.88 ± 0.83	36.84	16.0 ± 3.5
SubTrunc [29]	69.40 ± 1.00	38.65	16.0 ± 5.0
Oort [31]	21.14 ± 0.60	83.33	—

of five for DivFL and for SubTrunc, and none for Oort. The fairness guarantee is therefore the primary result and the information terms refine the convergence path rather than the final accuracy. Across $K \in \{1, 3, 5, 10\}$ the spread between variants stays between 0.46 and 1.58 percentage points with none leading consistently, so this is not specific to the headline.

Against the ported selectors the proposed method improves on DivFL by 1.33 and on SubTrunc by 1.81 percentage points in final accuracy, and it lowers their participation Gini coefficient by more than an order of magnitude. Against uniform random selection and against UnionFL the accuracy difference lies inside one standard deviation, so the three are treated as tied on that axis. UnionFL matches it on all three columns here, so the two separate elsewhere. Over the cohort sweep at $N = 30$ the proposed selector returns 6.67%, 0.00%, 1.33% and 0.67% at $K \in \{1, 3, 5, 10\}$, each the value (17) specifies, at a seed standard deviation of zero throughout. UnionFL returns 55.56%, 29.44%, 1.33% and 22.89% on the same runs, its seed standard deviation reaching 3.11. The agreement at $K = 5$ is a coincidence of one configuration and not a property of its objective. The two submodular methods concentrate participation heavily, above 36%, because their diversity objective is evaluated per round and carries no memory of which clients have already been served. The Oort result quantifies the argument of Section II-B. Transferring a utility-based FL selector into FD unmodified lowers final accuracy to 21.14 ± 0.60 and raises the Gini coefficient to 83.33%, and neither threshold is reached. The mechanism is in the selection record. A Gini coefficient of 83.33% over $N = 30$ at a seed standard deviation of zero is what appears when the same five clients take every slot of every round, which is what a utility ranking with no exploration term does once the loss ordering settles. The other 25 never contribute, so under (5) the distillation target is formed from a fifth of the pool, and the per-client accuracy standard deviation of 25.78 against 6.71 for uniform random follows. The row is read as a utility-only rule ported without a coverage mechanism rather than as a tuned comparison against Oort as published.

C. Sensitivity to the Selector Coefficients

The magnitude argument of Section IV-D predicts that any setting with $\alpha_u + \alpha_d < 1$ preserves the debt ordering. This is tested directly on a grid of $\alpha_u \in \{0, 0.1, 0.2, 0.3, 0.5, 0.8\}$ against $\alpha_d \in \{0, 0.05, 0.1, 0.2, 0.4\}$, giving thirty settings, each repeated over three seeds.

Twenty-eight of the thirty cells satisfy $\alpha_u + \alpha_d < 1$ and return a participation Gini coefficient of 1.33%, confirming the guarantee is independent of the admissible values. The other two, $(0.8, 0.2)$ and $(0.8, 0.4)$, fall outside the sufficient condition of Proposition 1 and both return 1.33% as well. Final accuracy ranges from 70.94% to 72.27%, so the entire grid spans 1.3 percentage points, which is comparable to the seed-to-seed spread at a single setting. The configuration used throughout this paper, $(\alpha_u, \alpha_d) = (0.3, 0.1)$, reaches 71.99% and sits within half a percentage point of the best cell. The selector therefore needs no per-configuration tuning.

D. Data Heterogeneity

The effect of the partition severity is examined by sweeping the Dirichlet concentration over $\alpha \in \{0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 5.0\}$ at $N = 30$, $K = 5$. Fig. 2(a) and (b) show the outcome. A separate IID run provides an upper anchor for the same pool and cohort.

Severe heterogeneity lowers accuracy for every selector alike. At $\alpha = 0.05$ the proposed selector reaches 26.74 ± 4.19 and uniform random reaches 27.07 ± 4.24, at $\alpha = 0.1$ 39.28 ± 2.55 and 39.14 ± 2.69, and at $\alpha = 5.0$ 84.67 ± 0.11 and 84.60 ± 0.18. Across the whole range the accuracy difference stays inside one standard deviation. The participation Gini coefficient is instead 1.33% for the proposed selector at every α tested, against 12.16% to 12.64% for uniform random, so the guarantee is invariant to heterogeneity as Proposition 1 predicts, since the debt term reads participation counts and not the data distribution. The two submodular baselines degrade on both axes as α falls, since their per-round diversity objective is less informative when clients hold few classes. At $\alpha = 0.01$ every selector collapses towards chance, with uniform random at 13.47 ± 1.33 and the proposed selector at 13.11 ± 1.20, since most clients hold a single class, yet the fairness separation persists at 1.33% against 12.16%. An IID partition anchors the other end at 85.68 ± 0.07 against 85.61 ± 0.09, with the same 1.33% against 12.49% separation, so the gap holds even with no heterogeneity at all.

E. Sparsity Sweep

Partial participation is characterized by sweeping $K \in \{1, 3, 5, 10\}$ at $N = 30$ over three seeds in Fig. 2(c) and (d).

At $K = 1$ the proposed selector reaches 60.61 ± 1.68 against 58.67 ± 2.28 for uniform random and at $K = 3$ it reaches 69.05 ± 0.94 against 68.04 ± 1.34, differences of 1.94 and 1.01 percentage points that are positive on every seed yet smaller than the seed variation. Three seeds place the smallest attainable two-sided p at 0.25, so the sparse regime is reported as a direction and not as a gain. At $K = 5$ and $K = 10$ the two are tied within one standard deviation. The participation Gini coefficient separates the two selectors at every K, and the separation widens as K falls, from 0.67% against 7.61% at

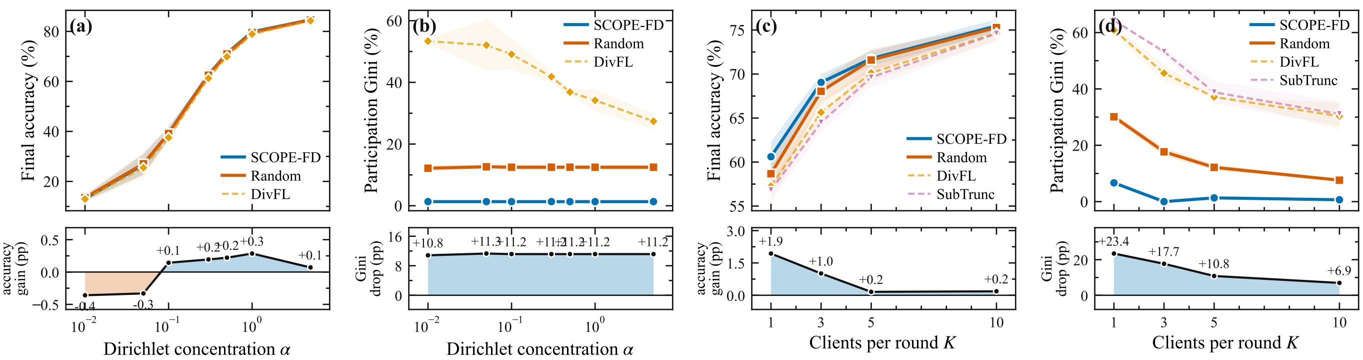

Fig. 2. Sensitivity of the selector on FMNIST at $N = 30$. Panels (a) and (b) sweep the Dirichlet concentration α at $K = 5$ and report final accuracy and the participation Gini coefficient. Panels (c) and (d) sweep the cohort size K over three seeds and report the same two quantities. The strip below each panel gives the margin of the proposed selector over uniform random selection, signed so that upward always favours the proposed selector. Shaded bands are one standard deviation over seeds.

$K = 10$ to 6.67% against 30.11% at $K = 1$. This is the regime the selector is designed for. As K approaches N, uniform random already covers most of the pool each round, so little imbalance is left to remove. Balanced participation reaches the clients and not only the counts. The standard deviation of final accuracy across the N clients falls at $K = 1$ from 13.94 ± 1.24 under uniform random to 9.11 ± 2.00 under the proposed selector and 8.95 ± 1.28 under the debt-only variant, in that direction on all three seeds. It is 7.93 against 7.47 at $K = 3$, 6.63 against 6.54 at $K = 5$ and 6.03 against 5.99 at $K = 10$. The effect tracks the imbalance it removes and is the empirical counterpart of Corollary 1.

F. Channel Robustness

The sensitivity to the wireless link is characterized at $N = 30$, $K = 5$ by sweeping the downlink SNR over $\{-30, -25, -20, -15, -10\}$ dB with uplink SNR fixed at -8 dB, using three seeds per point. Fig. 3(a) shows the result.

Accuracy improves smoothly with link quality, from 65.23 ± 1.39 at -30 dB to 67.39 ± 0.97 at -10 dB, a change of about two percentage points across the 20-dB range. The four variants track one another at every SNR, so the trend belongs to the FedTSKD substrate rather than the selection rule. The participation Gini coefficient is 1.33% at every SNR for all variants, so the guarantee is decoupled from the wireless layer, since the debt term reads participation counts that do not depend on the channel realization.

G. Public-Dataset Sensitivity

FD requires a shared public dataset, so sensitivity to that choice is examined along three axes, namely the identity of the set, its size and the label noise injected into it. Every cell in this study uses three seeds, and the baseline the cells are read against is 71.75 ± 0.81.

Identity has the largest effect. Replacing the held-out public set with MNIST lowers the accuracy of the proposed selector to 67.10 ± 1.05, reflecting the mismatch between the public inputs on which the consensus logits are formed and the private inputs the clients train on. That cell is also the disjoint-corpus control, since MNIST shares no input with the evaluation split. The ordering survives it, with the debt-only variant at 67.14 ± 0.95 and uniform random at 66.72 ± 0.89, so the accuracy level depends on the public set but the comparison does not. Size matters up to a threshold. Reducing the public set from 2000 to 500 samples costs about 2.5 percentage points, giving 69.22 ± 0.84, while reducing it further to 100 costs only 0.17 more, giving 69.05 ± 1.16. A few hundred public samples therefore suffice, which matters because every participant holds the set.

Label noise on the public set has no effect, and the reason is structural rather than empirical. The aggregation rule in (5) consumes public-set logits and never reads public-set labels, so noise applied to those labels cannot enter the pipeline. The measured values at noise levels 0.1 and 0.3 match the noise-free baseline of 71.75 ± 0.81 exactly. Across all of these cells the proposed selector degrades in step with uniform random rather than more gracefully, with the gap staying between 0.03 and 0.38 percentage points. The public-set choice therefore affects the pipeline as a whole rather than the selection rule. The participation Gini coefficient remains 1.33% throughout.

H. Privacy of the Label Histogram

Equations (12) and (14) read the per-client label histogram, which discloses the local class distribution to the server. Section IV-D names two remedies and both are evaluated here. The first applies a Laplace mechanism to the histogram before the selector sees it, swept over a privacy budget $\varepsilon_{\mathrm{dp}} \in \{0.1, 0.5, 1, 2, 5\}$. The second removes the disclosure entirely by replacing the histogram with a server-side surrogate formed from the softmax-averaged public-set logits that each client already submits under (5).

Neither remedy carries a measurable cost. Against 71.99 ± 0.63 for the undefended selector, the Laplace variants reach 72.01 ± 0.86 at $\varepsilon_{\mathrm{dp}} = 0.1$, 71.71 ± 0.72 at $\varepsilon_{\mathrm{dp}} = 0.5$, 71.97 ± 0.69 at $\varepsilon_{\mathrm{dp}} = 1$, 71.78 ± 0.80 at $\varepsilon_{\mathrm{dp}} = 2$, and 71.85 ± 0.73 at $\varepsilon_{\mathrm{dp}} = 5$. The surrogate reaches 71.93 ± 0.67. Every difference lies inside one standard deviation, including at the strongest noise setting. The participation Gini coefficient is 1.33% for all seven variants without exception. The ablation of Section VI-B explains why. The histogram enters the

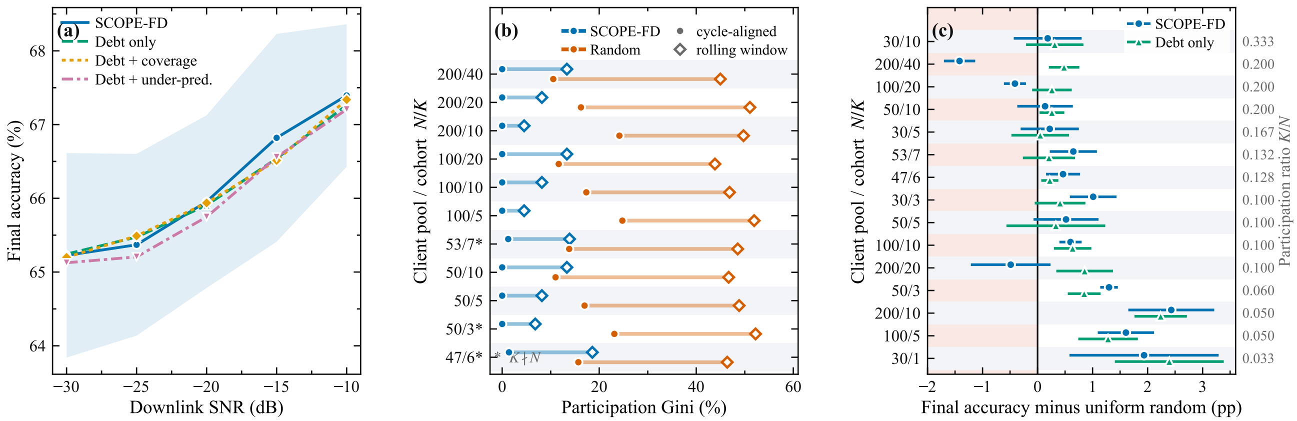

Fig. 3. Robustness of the participation guarantee on FMNIST. (a) Downlink signal-to-noise ratio sweep at $N = 30$, $K = 5$ over three seeds, where accuracy follows link quality while the participation Gini coefficient holds at 1.33% for all four score variants. (b) Participation fairness across pool and cohort sizes over five seeds, with filled circles the cycle-aligned Gini coefficient at $R = 100$, open diamonds the rolling-window value over an offset window, and stars marking $K \nmid N$. (c) Final accuracy relative to uniform random selection at $\alpha = 0.5$ over the fifteen matched configurations, ordered by participation ratio, pairing the complete score against the debt-only variant on the same seeds. The shaded band on the left of (c) marks the region where a selector trails uniform random. Bands and whiskers are one standard deviation over seeds.

score only through the under-prediction bonus and the class-coverage penalty, and those two terms were already shown not to change final accuracy, so perturbing their input cannot change it either. The participation-debt term that carries the fairness guarantee never reads the histogram at all, so the disclosure can be removed at no measured cost. This bounds the cost of the remedies rather than showing the value of the histogram, since a signal that does not move accuracy cannot be shown to matter by perturbing it.

I. Client Dropout and Bounded Staleness

Two departures from synchronous operation are examined. Under dropout each selected client fails to return its logits with probability p_{drop} and the server aggregates over whoever responded. Under bounded staleness a client may return its logits up to W rounds late, and the late contribution is aggregated with a staleness discount.

Under dropout the proposed selector holds both advantages together. At $p_{\text{drop}} = 0.1$ it reaches 71.16 ± 0.69 against 70.95 ± 1.02 for uniform random, at $p_{\text{drop}} = 0.2$ it reaches 70.41 ± 1.00 against 70.09 ± 1.04, and at $p_{\text{drop}} = 0.3$ it reaches 69.20 ± 1.03 against 69.16 ± 1.48. Accuracy falls for both as dropout rises, since fewer clients reach each round's consensus logits. The participation Gini coefficient moves from 0.80% to 1.88% across that range while uniform random moves from 13.42% to 15.45%. Dropout is the only perturbation examined here that moves the coefficient, because it alone changes which clients actually participated, and the debt term then rotates against the realized counts rather than the intended ones. Under bounded staleness the guarantee is untouched. At $W = 1$ the proposed selector reaches 71.71 ± 0.61 against 71.38 ± 1.20, and at $W = 2$ it reaches 71.54 ± 0.60 against 71.05 ± 1.18. The participation Gini coefficient is 1.33% at both windows, identical to the synchronous value, because a late arrival changes when a contribution is aggregated but not whether the client was selected.

J. Channel and Energy Awareness

The motivation of this paper rests on an mMIMO substrate, yet the score in (8) reads no channel or energy quantity. A fourth term $\alpha_c q_i$ was therefore added, reading the per-client channel quality and energy estimate, and evaluated under an explicit per-round energy budget.

The channel-aware variant is worse on both axes at every downlink SNR tested. It reaches 61.56 ± 3.31 at -30 dB, 62.48 ± 3.23 at -20 dB and 63.95 ± 3.04 at -10 dB, against 65.23 ± 1.39, 65.95 ± 1.17 and 67.39 ± 0.97 for the plain selector. The deficit of about 3.5 percentage points is stable across the sweep. Its participation Gini coefficient rises to 11.69% against 1.33%, its seed-to-seed spread roughly triples.

The mechanism is the energy budget rather than the extra score term. Instrumenting the selection loop shows it acting as a hard feasibility filter, removing any client whose energy exceeds the remaining allowance regardless of how far behind it is. The plain selector admits the full cohort of $K = 5$ every round. The channel-aware variant admits fewer in 89 of 100 rounds, the mean cohort falls to 3.56, the participation counts spread over a range of 15, and two of the thirty clients are never selected. Proposition 1 assumes every client remains selectable, and a filter acting on a criterion uncorrelated with debt breaks that assumption. Starving rounds of clients also thins each consensus logit, which accounts for the accuracy deficit. The proposed selector is therefore a fairness and data-coverage selector running on top of an mMIMO substrate rather than one that exploits channel state.

K. Client Scale, Divisibility and Window Alignment

The fairness guarantee is finally stressed along the three directions where its derivation is least immediate, namely a larger client pool, a cohort size that does not divide the pool, and an evaluation window that does not align with a participation cycle. Configurations with N up to 200 and K up to 40 are used, including the non-divisible cases $N = 47$

with $K = 6$ and $N = 53$ with $K = 7$. The largest cohort occupies 40 of the $N_{BS} = 64$ spatial dimensions available, so the sweep covers most of the range the rank condition of Section III-A admits. A rolling-window Gini coefficient is also computed over a window deliberately offset from the cycle length $\lceil N/K \rceil$, and Fig. 3(b) shows both.

Three results follow. First, the law holds at every pool size, and the zeros reached at the largest pool are an alignment of the horizon rather than an effect of the pool. At $N = 200$ with $K = 10$ the proposed selector reaches a cycle-aligned Gini coefficient of 0.00% against 24.16% for uniform random, and its accuracy is 51.27% against 48.84%. It stays at 0.00% for $K = 20$ and $K = 40$ at that pool, against 16.24% and 10.53%. Those zeros arise because KR is an exact multiple of N at $R = 100$, the vanishing condition of (17), so they carry no statement about pool size. Three rounds earlier the ordering reverses, at 2.63% for $N = 200$ with $K = 10$ against 0.86% at the headline pool, which is the oscillation Section V-A describes. Second, the non-divisible configurations behave as (17) predicts, giving 1.40% at $N = 47$ with $K = 6$ against 15.70% for uniform random, and 1.25% at $N = 53$ with $K = 7$ against 13.81%. Third, the advantage survives a measurement window that does not align with a cycle, since the rolling-window coefficient stays between 4.52% and 18.58% across all configurations while uniform random stays between 43.85% and 52.22%. The guarantee also holds at any horizon. At the headline configuration it falls from 3.33% at $R = 25$ to 1.33% at $R = 100$, matching (17) throughout and staying below the bound $N/(4KR)$, while uniform random falls only from 23.93% to 12.49%.

The one axis on which the proposed selector does not lead is accuracy at high participation ratios, and the campaign is large enough to make that statement quantitative. Fifteen configurations pair it against uniform random on matched seeds under otherwise identical settings, covering N from 30 to 200 and ratios K/N from 0.033 to 0.333. Fig. 3(c) collects them. The mean accuracy gap declines with the ratio, at a Spearman rank correlation of -0.815 against K/N, reaching $+2.43$ percentage points at $N = 200$ with $K = 10$ and -1.42 at $K = 40$. Twelve of the fifteen gaps are positive.

The three configurations in which it trails are $N = 100$ with $K = 20$, $N = 200$ with $K = 20$ and $N = 200$ with $K = 40$. All three use a cohort of at least twenty clients, and in all three the debt-only variant leads the complete score, by 0.67, 1.34 and 1.90 percentage points. Repeating the analysis for that variant makes the reason plain. Its mean gap over uniform random is positive in all fifteen configurations, from $+0.05$ to $+2.39$ percentage points, with the same decline against the ratio at -0.764. Several of these gaps are a small fraction of the seed spread, so the claim concerns the direction of the effect rather than its size. The fifteen configurations also share a seed set rather than being independent, so both correlations are reported as trends with no significance level.

Balanced rotation therefore costs no accuracy at any ratio tested, and what difference appears does so where participation is sparse. The two information terms are what turn the gap negative, once the cohort is large enough to cover the label space on its own. Cohort size and ratio are confounded here, but the three trailing cases all have $K \geq 20$ while $(30, 10)$ leads at a ratio of 0.333, so cohort size is the better-supported explanation. This agrees with the ablation of Section VI-B and confines any accuracy difference to the sparse regime. The per-round record shows the rotation directly. At $(N = 30, K = 10, R = 100)$ over three seeds, uniform random spreads the cumulative counts from 23 to 42 with a standard deviation of 4.53, while every client under the proposed selector sits at 33 or 34 with a standard deviation of 0.47. Every aligned window of $\lceil N/K \rceil = 3$ rounds partitions the pool exactly, in 33 of 33 windows on all three seeds and for both variants, which uniform random achieves in none. The two variants differ only in how they fill the cycle. The debt-only rule repeats three fixed cohorts, while the complete score draws on between 35 and 54 distinct ones.

L. *Results on Further Datasets*

Two further datasets test whether the conclusions depend on Fashion-MNIST, the first varying the image corpus and the second leaving images altogether.

On MNIST as the private set the ordering matches Fashion-MNIST closely. Over three seeds the debt-only variant reaches 90.88 ± 0.42, uniform random 90.58 ± 0.43 and the complete score 90.50 ± 0.56, all within one standard deviation, while DivFL reaches 88.48 ± 0.43 and SubTrunc 87.91 ± 0.50. The participation Gini coefficient is again 1.33% against 12.16% for uniform random and above 36% for the submodular pair.

To test a domain outside images the selector is run on the FSDD, a ten-class spoken-digit corpus represented as log-spectrograms and paired with the `AudioCNN` model. Two settings differ from the image studies, both forced by corpus size. It gives roughly ninety training samples per client, about twenty times fewer than Fashion-MNIST, so local epochs rise from one to three, and its held-out split holds only three hundred samples, so the public set is cut to one hundred and fifty. Over five seeds the proposed selector reaches 46.16 ± 1.69, the debt-only variant 45.54 ± 1.17, uniform random 44.99 ± 1.59, SubTrunc 40.66 ± 1.48 and DivFL 40.53 ± 1.49, with the participation Gini coefficient at 1.33% against 12.49% for uniform random and about 45% for the two submodular methods. The proposed selector leads uniform random on all five seeds by between 0.51 and 1.70 percentage points, and both submodular methods on every seed by at least four. Absolute accuracy is far below the image results because of the small per-client sample count, so the ordering rather than the level is informative. That ordering reproduces the image result, including the tie between the complete score and the debt-only variant, which trails by 0.61 percentage points at a two-sided paired p of 0.188. Across the three datasets the fairness separation is identical, because it depends only on the participation arithmetic, and the accuracy ordering is preserved throughout. What changes is the absolute level, which reflects the difficulty of the task rather than the selection rule.

M. *Invariance of the Fairness Guarantee*

The campaign spans seventeen experiment families, which allows (17) to be tested as a prediction rather than reported one setting at a time.

For every run N, K and R are known before it starts, so (17) fixes the participation Gini coefficient in advance. Across the campaign it is correct in 459 of 459 cases, which checks an analytic identity against a deterministic procedure rather than testing a hypothesis, covering 276 runs of the proposed selector and 183 of the debt-only variant, with no deviation in any run. Held out are the eighteen dropout runs, where a selected client may fail to return and the realized participation is therefore not the intended participation. The measured value is also seed-independent, with a standard deviation of exactly zero at every configuration. The law holds across every condition reported in Sections VI-C to VI-L, covering all three datasets, all seven heterogeneity levels, the SNR sweep, every privacy mechanism, both staleness windows, every public-set variation and all thirty coefficient cells. The value 1.33% is returned in 313 of these runs across fifteen of the seventeen families, and every other run returns what its triple specifies.

This follows from the structure of the score, since the right-hand side of (17) contains only N, K and R. Fairness can therefore be predicted from the deployment parameters instead of being re-validated for each condition, which a penalty-based or stochastic fairness objective cannot offer. The one condition under which it degrades, a hard resource constraint that renders clients unselectable, is characterized in Section VI-J. Dropout is the milder form of the same effect, and the coefficient then varies from run to run between 0.42% and 2.12%.

VII. Conclusion and Future Work

This paper proposed SCOPE-FD, a deterministic fairness-first client selector for federated distillation over mMIMO networks under partial participation, built on the FedTSKD substrate of [12]. The selector shapes the over-round composition of the client subset through a debt-dominated three-term score with fixed non-negative coefficients. Its central result is that fairness becomes predictable rather than merely good. The cumulative participation Gini coefficient is $m(N-m)/(NKR)$ with $m = KR \bmod N$, so it follows from the pool size, the cohort size and the horizon alone, and that value was returned in all 459 selector runs of the study free of client dropout. At the headline configuration it falls from 12.49% under uniform random selection to 1.33% while accuracy is matched at 71.21 ± 0.99 against 70.99 ± 1.38. Composed with the per-client bound of [12], the same guarantee yields a rate of $\mathcal{O}(N/(KR))$ in communication rounds that holds for every client at every finite horizon and adds no assumption to those of [12]. Three limits bound the result. Accuracy is preserved rather than improved, with the sparse-regime differences inside the seed variation and the comparison turning against the complete score once the cohort reaches twenty clients, where the debt-only variant should be preferred. The label-histogram disclosure that the score requires is removable at no measured cost. A hard energy budget instead breaks the guarantee, because filtering clients on a criterion uncorrelated with participation debt violates the assumption of Proposition 1 that every client remains selectable. Two directions are left open, namely admitting channel and energy signals into the score without losing the guarantee, and non-image domains on larger corpora.

References

[1] H. B. McMahan *et al.*, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.

[2] K. Borazjani *et al.*, "Redefining non-iid data in federated learning for computer vision tasks: Migrating from labels to embeddings for task-specific data distributions," *IEEE Trans. Artif. Intell.*, 2026.

[3] B. Rao *et al.*, "Privacy inference attack and defense in centralized and federated learning: A comprehensive survey," *IEEE Trans. Artif. Intell.*, vol. 6, no. 2, pp. 333–353, 2025.

[4] W. Huang *et al.*, "Federated learning for generalization, robustness, fairness: A survey and benchmark," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9387–9406, Dec. 2024.

[5] E. Jeong *et al.*, "Communication-efficient on-device machine learning: Federated distillation and augmentation under non-IID private data," in *Proc. NeurIPS Workshop MLPCD*, 2018.

[6] S. Itahara *et al.*, "Distillation-based semi-supervised federated learning for communication-efficient collaborative training with non-IID private data," *IEEE Trans. Mobile Comput.*, vol. 22, no. 1, pp. 191–205, Jan. 2023.

[7] F. Sattler *et al.*, "CFD: Communication-efficient federated distillation via soft-label quantization and delta coding," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 4, pp. 2025–2038, 2022.

[8] ——, "FedAUX: Leveraging unlabeled auxiliary data in federated learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 5531–5543, Sep. 2023.

[9] Y.-H. Chan and E. C.-H. Ngai, "FedHe: Heterogeneous models and communication-efficient federated learning," in *Proc. Int. Conf. Mobility, Sens. Netw. (MSN)*, 2021, pp. 207–214.

[10] Y. J. Cho *et al.*, "Communication-efficient and model-heterogeneous personalized federated learning via clustered knowledge transfer," *IEEE J. Sel. Topics Signal Process.*, vol. 17, no. 1, pp. 234–247, Jan. 2023.

[11] Z. Wu *et al.*, "FedICT: Federated multi-task distillation for multi-access edge computing," *IEEE Trans. Parallel Distrib. Syst.*, vol. 35, no. 6, pp. 1107–1121, Jun. 2024.

[12] Y. Mu, N. Garg, and T. Ratnarajah, "Federated distillation in massive MIMO networks: Dynamic training, convergence analysis, and communication channel-aware learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 4, pp. 1535–1550, Aug. 2024.

[13] J.-H. Ahn, O. Simeone, and J. Kang, "Wireless federated distillation for distributed edge learning with heterogeneous data," in *Proc. IEEE PIMRC*, 2019, pp. 1–6.

[14] M. M. Amiri and D. Gündüz, "Federated learning over wireless fading channels," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3546–3557, May 2020.

[15] T. T. Vu *et al.*, "Cell-free massive MIMO for wireless federated learning," *IEEE Trans. Wireless Commun.*, vol. 19, no. 10, pp. 6377–6392, Oct. 2020.

[16] D. Thakur *et al.*, "GRACE-FL: Green resource-aware communication-efficient federated learning," *IEEE Trans. Artif. Intell.*, vol. 7, no. 6, pp. 3221–3236, 2026.

[17] T. Nishio and R. Yonetani, "Client selection for federated learning with heterogeneous resources in mobile edge," in *Proc. IEEE ICC*, 2019, pp. 1–7.

[18] H. H. Yang *et al.*, "Scheduling policies for federated learning in wireless networks," *IEEE Trans. Commun.*, vol. 68, no. 1, pp. 317–333, Jan. 2020.

[19] J. Ren *et al.*, "Scheduling for cellular federated edge learning with importance and channel awareness," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7690–7703, Nov. 2020.

[20] W. Shi *et al.*, "Joint device scheduling and resource allocation for latency constrained wireless federated learning," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 453–467, Jan. 2021.

[21] W. Xia *et al.*, "Multi-armed bandit-based client scheduling for federated learning," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7108–7123, Nov. 2020.

[22] Y. J. Cho *et al.*, "Bandit-based communication-efficient client selection strategies for federated learning," in *Proc. Asilomar Conf. Signals, Syst., Comput.*, 2020, pp. 1066–1069.

[23] R. Balakrishnan *et al.*, "Diverse client selection for federated learning via submodular maximization," in *Proc. ICLR*, 2022.

[24] L. Liu *et al.*, "Communication-efficient federated distillation with active data sampling," in *Proc. IEEE ICC*, 2022, pp. 201–206.

[25] S. Wang *et al.*, "Adaptive federated learning in resource constrained edge computing systems," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1205–1221, Jun. 2019.

[26] K. Kazemi, M. H. Badiei, and H. Kebriaei, "Robust peer-to-peer federated learning with deep reinforcement learning based client selection against data poisoning attacks," *IEEE Trans. Artif. Intell.*, vol. 7, no. 1, pp. 262–271, 2026.

[27] T. Li *et al.*, "Fair resource allocation in federated learning," in *Proc. ICLR*, 2020.

[28] S. Arisdakessian *et al.*, "Towards vox populi in federated learning: A fair and inclusive client selection framework," *IEEE Trans. Artif. Intell.*, vol. 7, no. 4, pp. 1997–2011, 2026.

[29] A. C. Castillo J. *et al.*, "Equitable client selection in federated learning via truncated submodular maximization," in *Proc. IEEE CDC*, 2024, pp. 5496–5502.

[30] ——, "Submodular maximization approaches for equitable client selection in federated learning," arXiv:2408.13683, 2024.

[31] F. Lai *et al.*, "Oort: Efficient federated learning via guided participant selection," in *Proc. USENIX OSDI*, 2021, pp. 19–35.

[32] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," in *Proc. NIPS Deep Learning Workshop*, 2014.

[33] M. Yurochkin *et al.*, "Bayesian nonparametric federated learning of neural networks," in *Proc. ICML*, 2019, pp. 7252–7261.